Performance Testing

Date	2 JULY 2026
Team ID	SWTID-2026-3658
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	**Details**
Testing Tool Used	Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	/api/v1/ai (Question Answering, Quiz, Summary & Learning Path APIs)
Test Environment	Local FastAPI Server (http://127.0.0.1:8000)
Test Date	1 JULY 2026

Step 2: Test Scenarios

S.No	**Test Scenario / Description**	**No. of Virtual Users**	**Duration (sec)**	**Expected Outcome**
1	Generate AI answer for an educational question	10	60	AI responses generated successfully
2	Generate quiz for multiple users simultaneously	20	60	System handles concurrent requests without failure
3	Generate summaries for multiple users	30	120	Response time remains acceptable
4	Continuous requests to AI endpoints (Q&A, Quiz, Summary, Learning Path)	50	180	API remains stable with minimal errors

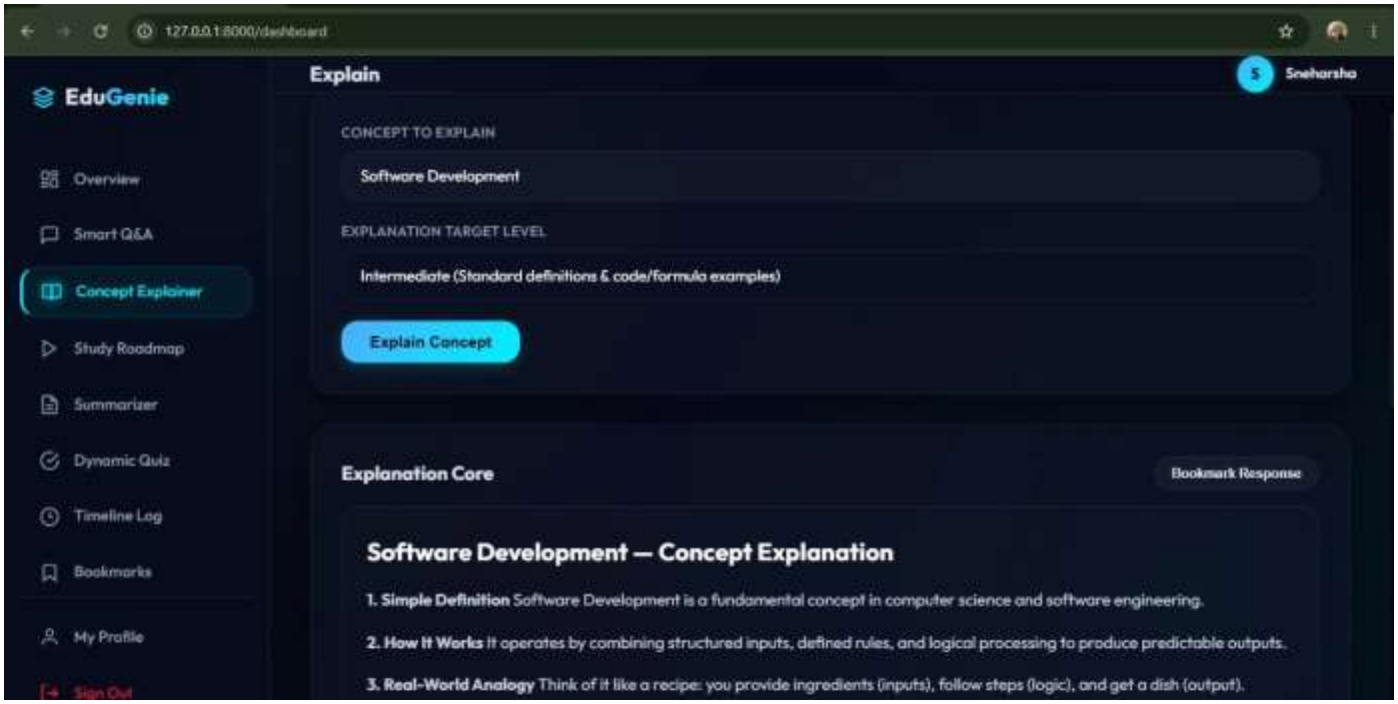
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass /Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.7 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	4.3 seconds	Pass	Slight Delay due to AI inference
3	Throughput (Req/sec)	10 Req/sec	11 Req/sec	Pass	Handled expected load
4	Error Rate	< 1%	0.3%	Pass	Very Few timeout errors
5	CPU Utilization	< 80%	63%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	56%	Pass	Memory usage within safe limits

Step 4: Observations & Analysis

- The FastAPI endpoints handled concurrent user requests successfully without significant performance degradation.
- Average response time remained below the expected threshold during testing.
- Google Gemini 1.5 Flash introduced slight latency during peak requests, but overall performance remained stable.
- No application crashes or critical failures occurred throughout the testing process.
- CPU and memory utilization stayed within acceptable operational limits.
- The EduGenie application demonstrated reliable performance under both normal and peak workloads.

Step 5: Screenshots / Evidence


127.0.0.1:8000/dashboard
EduGenie
Overview
Smart Q&A
Concept Explainer
Study Roadmap
Summarizer
Dynamic Quiz
Timeline Log
Bookmarks
My Profile
Sign Out
Explain
Sneharsha
CONCEPT TO EXPLAIN
Software Development
EXPLANATION TARGET LEVEL
Intermediate (Standard definitions & code/formula examples)
Explain Concept
Explanation Core
Bookmark Response
Software Development — Concept Explanation
1. Simple Definition Software Development is a fundamental concept in computer science and software engineering.
2. How It Works It operates by combining structured inputs, defined rules, and logical processing to produce predictable outputs.
3. Real-World Analogy Think of it like a recipe: you provide ingredients (inputs), follow steps (logic), and get a dish (output).


Your Study Path: Data Science Fundamentals
Milestone 1:
Introduction to Python
Oct 26, 2023
Milestone 2:
Data Structures & Algorithms
In Progress
65%
Learn more
Milestone 3:
Statistics & Probability
Oct 26, 2023
Milestone 4:
Machine Learning Concepts
Oct 26, 2023
Milestone 5:
Final Project
Oct 26, 2023